



Synthesis, kinetics and characterization of environment friendly waterborne acrylate copolymers

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ARTICLE INFO

Article history:

Received 16 March 2020

Received in revised form 7 April 2020

Accepted 10 April 2020

Available online 30 April 2020

Keywords:

Acrylate

Coating

Copolymer

Kinetics

EHA

ABSTRACT

The copolymers consisting of acrylic and methacrylic esters have achieved prime importance due to their versatile environment friendly applications. Copolymerization of Methyl methacrylate (MMA) with suitable vinyl monomers such as alkyl acrylate and methacrylate has been carried out for modifying the properties of the polymer. In current research synthesis of waterborne copolymer of MMA and 2-Ethyl hexyl acrylate (2-EHA) carried out by ecofriendly emulsion polymerization technique and the kinetics of polymerization, characterizations of the copolymers have been studied. Synthesized copolymers were characterized by FTIR, NMR, DSC, and TGA. The kinetics results of the MMA-co-EHA copolymer was compared with the previously synthesized copolymer MMA with Butyl acrylate (BA). On comparing these results, it was found that EHA gives higher conversion with respect to BA under the same reaction conditions. Out of all the prepared latexes some are suitable for binder purposes in environment friendly water borne coating applications.

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Selection and peer-review under responsibility of the scientific committee of the 3rd International Conference on Science and Engineering of Materials.

1. Introduction

Copolymer synthesis is important in several areas due to the wide range of properties that can be achieved from the combination of different monomers. The copolymers consisting of acrylic and methacrylic esters have achieved prime importance due to their versatile applications [1–3] in medicine, pharmaceuticals, paints, polymer activities and rubbers. Copolymerization of methyl methacrylate (MMA) with suitable vinyl monomers such as alkyl acrylate and methacrylate has been carried out for modifying the properties of the polymer. The length of the alkyl group in these alkyl acrylates has pronounced effects on the flexibility of methacrylate polymers. The increase in the side chain length of the esters derived from straight chain alcohol usually softens the polymer.

Acrylic polymers cover a wide range of physical properties. Depending on the monomers selected and the method of polymerization the acrylate esters produce internal plasticization of the polymers resulting in improved toughness, increased tensile strength, increased tear and impact strengths and improved process ability. The selection of the monomer and monomer ratios in the copolymerization play a vital role in designing the copoly-

mers for the use in the coatings and paint applications. 2-Ethyl hexylacrylate (EHA) is a very useful monomer in number of industrial applications. The poly (ethyl hexyl acrylate) (PEHA) has very low glass transition temperature ($T_g = -65^\circ\text{C}$) and also it has excellent film formation property [4]. The copolymers of PEHA with other acrylates, methacrylates, esters, vinyl acetate, vinyl chloride, etc. find applications in adhesives, coatings, leather, plasticizers, chemical intermediates, textiles and in many plastics [5,6].

n-Butyl acrylate (BA), 2-ethyl hexyl acrylate (EHA), and MMA monomers specifically are used in adhesive production and coating applications as binders in household paints [7,8] largely because their different glass transition temperatures allow for the control the application properties [9,10]. The T_g of PMMA and PBA are 100°C and -49°C respectively.

The kinetics of copolymerization of MMA with BA and their characterization has been discussed in previous study [11]. The aim of this research to prepare an environment friendly latex and to study the kinetics of copolymerization of MMA and 2-EHA. The characterizations of the copolymers will be studied to see the effect of higher acrylate (2-EHA) on the kinetics of polymerization. The structural characterization of the copolymer was performed by FTIR and ^1H NMR. The reactivity ratios in the